

EXPERIENCE

Moreland Connect

Senior Software Engineer

Jun 2025 - Present

Twinsburg, OH

Software consultancy - architected and delivered production systems across concurrent client engagements

- Rebuilt an IoT sensor platform's failing legacy portal end-to-end (React/Spring Boot, PostgreSQL, Timescale sustaining 55 writes/sec); upgrading 20+ MQTT-to-SQS pub/sub services connecting gateways to the portal
- Technical lead for a 3-engineer pharmaceutical epidemiology and market forecasting platform, designing system architecture and owning the agent microservice (Python/FastAPI, Anthropic, Valyu)
- Architected a serverless special-orders platform (Lambda, EventBridge, SQS/SNS) consolidating 20+ legacy applications for a billion-dollar retail client; wrote the initial backend, CDK, and enterprise system integrations
- Led technical design as architect for a 2-engineer rep portal team; own the Azure infrastructure and SAP integration (OData, BTP APIs, Event Mesh via iFlows) powering sales quotes and orders

Cleveland VA Medical Center

Data Scientist

Sept 2023 - May 2025

Cleveland, OH

- Developed multi-modal predictive models for device-adherence prediction; found predictive power varied across the population and implemented rule and tree-based subgroup discovery methods for 1.5x higher accuracy
- Architected Azure data pipelines consolidating unstructured clinical notes, time-series device telemetry, and patient metadata from disparate source systems into model ready datasets

Materials Data Science for Stockpile Stewardship (MDS3)

Applied Machine Learning Researcher

Jan 2022 - Dec 2024

Cleveland, OH

- Built CI/CD pipelines, Docker/Singularity containers, Slurm automation, and Open OnDemand web applications for HPC infrastructure serving 100+ daily active researchers on a 2.5-petabyte Hadoop cluster
- Refactored a data pipeline fusing acoustic, pyrometer, and CT sensor streams into a unified data lake, cutting processing time 10x+ and storage footprint 10x across 100+ GB per-sample datasets
- Developed an active learning and segmentation framework for scientific imaging achieving a 0.94 F1-score while labeling only 0.03% of the 342K-image dataset and detecting 5M+ defects

NASA Glenn Research Center

Applied Machine Learning

January 2023 - August 2023

Cleveland, OH

- Extended domain pretrained encoders (MicroNet) for few-shot segmentation, image regression, and self-supervised learning tasks; built VAE-based generative networks using MicroNet for synthetic microstructure texture synthesis
- Benchmarked MicroNet against ImageNet across 6 CNN architectures for fracture-strain regression on weld fractography images, improving low-data prediction accuracy by 23%
- Designed automated hyperparameter infrastructure (Slurm and Optuna) to schedule training runs on shared cluster

Emerson (RIGID Tools)

Data Scientist

Aug 2021 - May 2022

Elyria, OH

- Developed and optimized SQL ETL pipelines using Azure to reduce historical inventory queries by 40%
- Implemented real-time anomaly detection in drain inspection camera feeds using computer vision
- Improved email support case response times 35% by building and evaluating BERT and GPT-3 classifiers

ADDITIONAL EXPERIENCE (full dates and details available on request)

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| • LLNL (2024): sensor fusion, graph neural nets | • Progressive (2020): mobile UI/UX, Android |
| • INSIGHT2PROFIT (2021): full-stack pricing app | • GE Lighting (2020): IoT data infrastructure |
| • NBC Universal (2021): ETL automation, PowerBI | • Trimble (2019): ERP/CRM governance |

SKILLS

Languages: Python, TypeScript, SQL, Java, Go, C#, R

Frameworks: React, React Native, Next.js, Spring Boot, ASP.NET Core, PyTorch, TensorFlow, Apache Spark

Infrastructure: AWS, Azure, Docker, Railway, Vercel, GCP, PostgreSQL, Redis, Prefect, Slurm

EDUCATION

Case Western Reserve University

Master of Science in Computer Science - **GPA 4.0/4.0**

Bachelor of Arts in Computer Science and Cognitive Science - GPA 3.8/4.0

Cleveland, OH

Dec 2024

Jan 2022